AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

FACULTY OF SCIENCE AND TECHNOLOGY

CSC2210: OBJECT ORIENTED PROGRAMMING 2

FALL 2024-2025

Section: Q

Group No: 03

Project Report On

Blood & Organ Donation System

Supervised By

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Submitted By

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Introduction:

Welcome to the Project **Blood & Organ Donation System**, this project aims to develop a comprehensive Blood and Organ Donation System to streamline and enhance the efficiency of the donation process. The system will facilitate the registration of donors, manage donor and recipient data, track donations, and support healthcare providers in ensuring the equitable and timely distribution of blood and organs. By leveraging advanced technologies, the system seeks to optimize donor-recipient matching, reduce waiting times, and ultimately improve patient outcomes.

Scenario Description:

The **Blood & Organ Donation System** is designed to streamline the donation process, connecting donors and recipients efficiently. The platform manages user registration, donation records, matching processes, and additional services such as emergency alerts and family matching, all aimed at saving lives and enhancing user experience.

A user registers on the system by providing essential information, including their name, contact details, user type (Blood Donor, Organ Donor, Blood Recipient, or Organ Recipient), and other personal details. The user can update or delete his/her profile. This profile helps manage future donations and matches, allowing users to participate as both donors and recipients if needed.

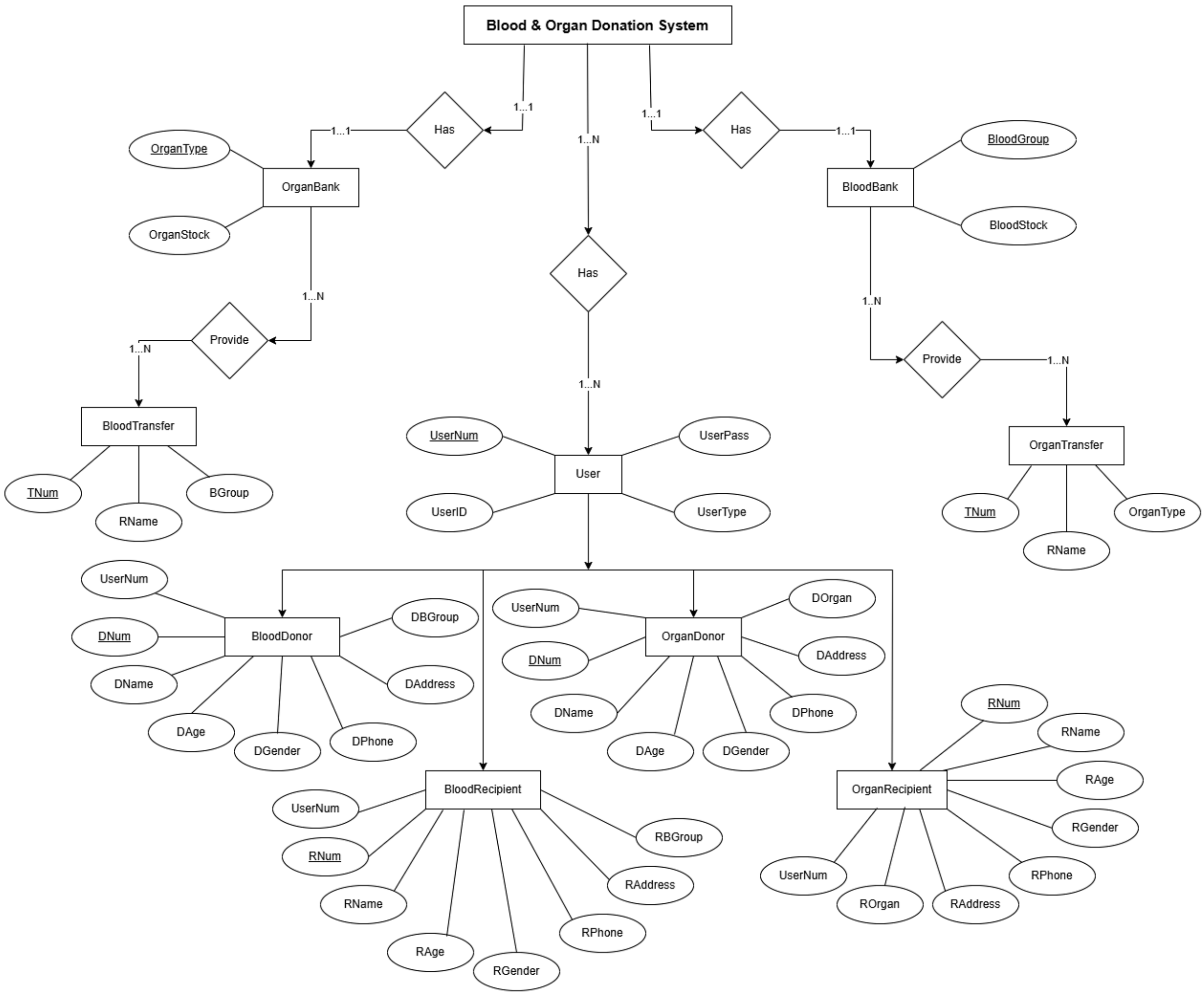
A blood donor expresses the intent to donate by registering through the system. The system records the donation details, including blood type, updates the donor's history, and marks their availability status. This information is stored in the Blood Bank, ensuring that the stock is accurately updated and ready for future use.

A recipient in need of a blood transfusion submits a request. The system processes the request by searching the Blood Bank for compatible blood types. Once a match is found, the system initiates the blood transfer process.

Similarly, an organ donor registers their intent to donate a specific organ. The system records the organ type and updates the Organ Bank accordingly. When an organ recipient submits a request, the system searches for compatible organ types, considering factors like geographical proximity. Upon identifying a match, the system facilitates the organ transfer and notifies both parties.

This scenario outlines the seamless integration of features in the **Blood & Organ Donation System**, ensuring efficient and effective donation processes to support healthcare needs and save lives.

ER Diagram:



Normalization:

Let’s assume we start with a single unnormalized table that includes all the information:

1. UserNum
2. UserID
3. UserPass
4. UserType
5. DNum
6. DName
7. DAge
8. DGender
9. DPhone
10. DAddress
11. DBGroup

12. RNum
13. RName
14. RAge
15. RGender
16. RPhone
17. RAddress
18. RBGroup
19. BGroup
20. BStock
21. TNum
22. DOrgan
23. ROrgan
24. OrganType
25. OrganStock

Separate into Related Entities (1NF)

1. User Table:
    i. UserNum
    ii. UserID
    iii. UserPass
    iv. UserType
2. BloodDonor Table:
    i. UserNum
    ii. DNum
    iii. DName
    iv. DAge
    v. DGender
    vi. DPhone
    vii. DAddress
    viii. DBGroup
3. BloodRecipient Table:
    i. UserNum
    ii. RNum
    iii. RName
    iv. RAge
    v. RGender
    vi. RPhone
    vii. RAddress

viii. RBGroup

4. Blood Table:
    i. BGroup
    ii. Bstock
5. BloodTransfer Table:
    i. TNum
    ii. RName
    iii. BGroup
6. OrganDonor Table:
    i. UserNum
    ii. DNum
    iii. DName
    iv. DAge
    v. DGender
    vi. DPhone
    vii. DAddress
    viii. DOrgan
7. OrganRecipient Table:
    i. UserNum
    ii. RNum
    iii. RName
    iv. RAge
    v. RGender
    vi. RPhone
    vii. RAddress
    viii. ROrgan
8. Organ Table:
    i. OrganType
    ii. OrganStock
9. OrganTransfer Table:
    i. TNum
    ii. RName
    iii. OrganType

Eliminate Partial Dependencies (2NF)

Not needed

Eliminate Transitive Dependencies (3NF)

Normalize Information into Specific Tables

1. BloodDonor Table:
    i. DNum (UserNum)
    ii. DName
    iii. DAge
    iv. DGender
    v. DPhone
    vi. DAddress
    vii. DBGroup
2. BloodRecipient Table:
    i. RNum (UserNum)
    ii. RName
    iii. RAge
    iv. RGender
    v. RPhone
    vi. RAddress
    vii. RBGroup
3. OrganDonor Table:
    i. DNum (UserNum)
    ii. DName
    iii. DAge
    iv. DGender
    v. DPhone
    vi. DAddress
    vii. DOrgan
4. OrganRecipient Table:
    i. RNum (UserNum)
    ii. RName
    iii. RAge
    iv. RGender
    v. RPhone
    vi. RAddress
    vii. ROrgan

Final Table and Optimized Table:

Database Name: BloodBondDB

1. **Table Name:** UserTbl

   Columns:

   i. UserID (PRIMARY KEY)
   ii. UserName
   iii. UserPass
   iv. UserType

Query:

SELECT TOP 1000 [UserNum]
,[UserID]
,[UserPass]
,[UserType]
FROM [BloodBondDb].[dbo].[UserTbl]

Table:

	UserNum	UserID	UserPass	UserType
1	101	Samiha	1234	User
2	102	Munira	3456	User
3	103	Mrini	1111	Employee

2. **Table Name:** DonorTbl

   Columns:

   i. DNum (PRIMARY KEY)
   ii. DName
   iii. DAge
   iv. DGender
   v. DPhone
   vi. DAddress
   vii. DBGroup

Query:

SELECT TOP 1000 [DNum]
,[DName]
,[DAge]
,[DGender]
,[DPhone]
,[DAddress]
,[DBGroup]

FROM [BloodBondDb].[dbo].[DonorTbl]

Table:

	DNum	DName	DAge	DGender	DPhone	DAddress	DBGroup
1	2001	Munira	21	Female	123456	Uttara	O+
2	2002	Mishu	22	Female	111111	Dhaka	B+
3	2003	Nusrat	21	Female	100000	Tongi	O+
4	2004	Samia	20	Female	126789	Dhaka	B+
5	2005	Mrini	23	Female	100	Uttara	B+
6	2006	Fahim	28	Male	222222	Kuril	AB+

3. **Table Name:** RecipientTbl

   Columns:

   i. RNum (PRIMARY KEY)
   ii. RName
   iii. RAge
   iv. RGender
   v. RPhone
   vi. RAddress
   vii. RBGroup

Query:

```
SELECT TOP 1000 [RNum]
      ,[RName]
      ,[RAge]
      ,[RGender]
      ,[RPhone]
      ,[RAddress]
      ,[RBGroup]
  FROM [BloodBondDb].[dbo].[RecipientTbl]
```

Table:

	RNum	RName	RAge	RGender	RPhone	RAddress	RBGroup
1	1001	Samiha	18	Female	123456	Dhaka	AB+
2	1002	Nisa	20	Female	333221	Tongi	B+
3	1003	Ayman	12	Male	111222	Bashundhara	O+
4	1004	Tasfia	22	Female	111111	Uttara	AB-

4. **Table Name:** BloodTbl

   Columns:

   i. BGroup (PRIMARY KEY)
   ii. BStock

Query:

```
SELECT TOP 1000 [BGroup]
      ,[BStock]
  FROM [BloodBondDb].[dbo].[BloodTbl]
```

Table:

	BGroup	BStock
1	A+	0
2	A-	0
3	AB+	1
4	AB-	3
5	B+	5
6	B-	0
7	O+	2
8	O-	0

5. **Table Name:** BloodTransferTbl

   Columns:

   i. TNum (PRIMARY KEY)
   ii. RName
   iii. BGroup

Query:

```
SELECT TOP 1000 [TNum]
      ,[RName]
      ,[BGroup]
  FROM [BloodBondDb].[dbo].[BloodTransferTbl]
```

Table:

	TNum	RName	BGroup
1	6001	Munira	O+
2	6002	Samia	B+
3	6003	Nisa	O+

6. **Table Name:** OrganDonorTbl

   Columns:

   i. DNum (PRIMARY KEY)
   ii. DName
   iii. DAge
   iv. DGender
   v. DPhone
   vi. DAddress
   vii. DOrgan

Query:

SELECT TOP 1000 [DNum]

,[DName]

,[DAge]

,[DGender]

,[DPhone]

,[DAddress]

,[DOrgan]

FROM [BloodBondDb].[dbo].[OrganDonorTbl]

Table:

	DNum	DName	DAge	DGender	DPhone	DAddress	DOrgan
1	5001	John Doe	35	Male	987653534	123, Elm Street	Kidney
2	5002	Jane Smith	29	Female	456778889	456, Oak Avenue	Liver
3	5003	Sam Brown	42	Male	111111111	789, Pine Road	Heart
4	5004	Samiha	22	Female	1233456	Dhaka	Eyes
5	5005	Emily White	39	Female	23647457	321, Maple Street	Lung
6	5006	Cheris Green	45	Male	66789230	654, Cedar Lane	Kidney
7	5007	Michael John	23	Male	55568789	123, Elm Street	Bone Marrow

7. **Table Name:** OrganRecipientTbl

   Columns:

   i. RNum (PRIMARY KEY)
   ii. RName
   iii. RAge
   iv. RGender
   v. RPhone
   vi. RAddress
   vii. ROrgan

Query:

SELECT TOP 1000 [RNum]

,[RName]

,[RAge]

,[RGender]

,[RPhone]

,[RAddress]

,[ROrgan]

FROM [BloodBondDb].[dbo].[OrganRecipientTbl]

Table:

	RNum	RName	RAge	RGender	RPhone	RAddress	ROrgan
1	4001	Michael Davis	39	Male	55551234	654, Cedar Lane	Liver
2	4002	Emma White	33	Female	55521987	321, Birch Street	Heart
3	4003	David Black	27	Male	55598876	123, Oak Street	Lung
4	4004	Sarah Lewis	30	Female	55567879	432, Elm Street	Kidney
5	4005	Alice Johnson	41	Male	55523478	987, Maple Boulevard	Kidney
6	4006	Alina	28	Female	55583726	321, Birch Street	Liver

8. **Table Name:** OrganTbl

   Columns:

   i. OrganType (PRIMARY KEY)

   ii. OrganStock

Query:

SELECT TOP 1000 [OrganType]

,[OrganStock]

FROM [BloodBondDb].[dbo].[OrganTbl]

Table:

	OrganType	OrganStock
1	Bone Marrow	1
2	Eyes	17
3	Heart	8
4	Intestines	0
5	Kidney	15
6	Liver	25
7	Lungs	15
8	Pancreas	0

9. **Table Name:** OrganTransferTbl

   Columns:

   i. TNum (PRIMARY KEY)

   ii. RName

   iii. OrganType

Query:

SELECT TOP 1000 [TNum]

,[RName]

,[OrganType]

FROM [BloodBondDb].[dbo].[OrganTransferTbl]

Table:

	TNum	RName	OrganType
1	3001	Alice Johnson	Kidney
2	3002	Michael Davis	Liver
3	3003	Emma White	Heart

User Story:

User Story of Admin:

1. UserName
2. Password

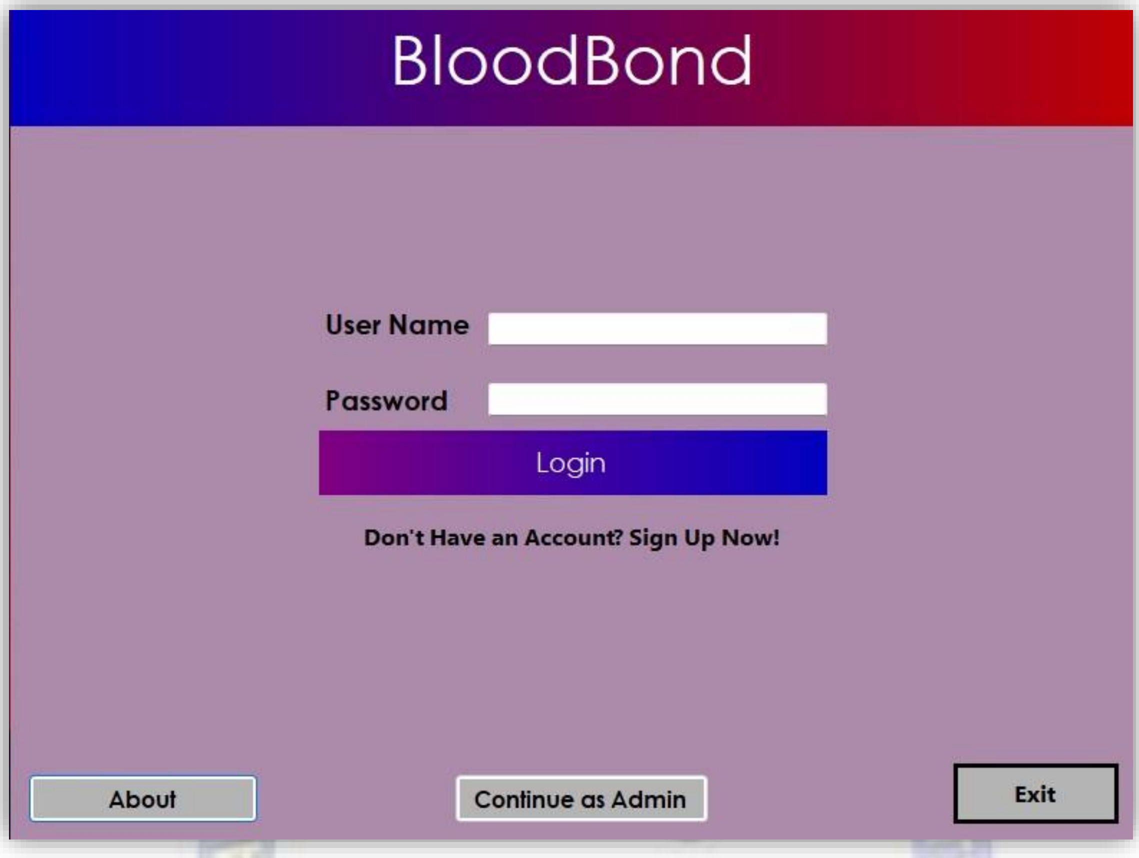
User Story of Donor:

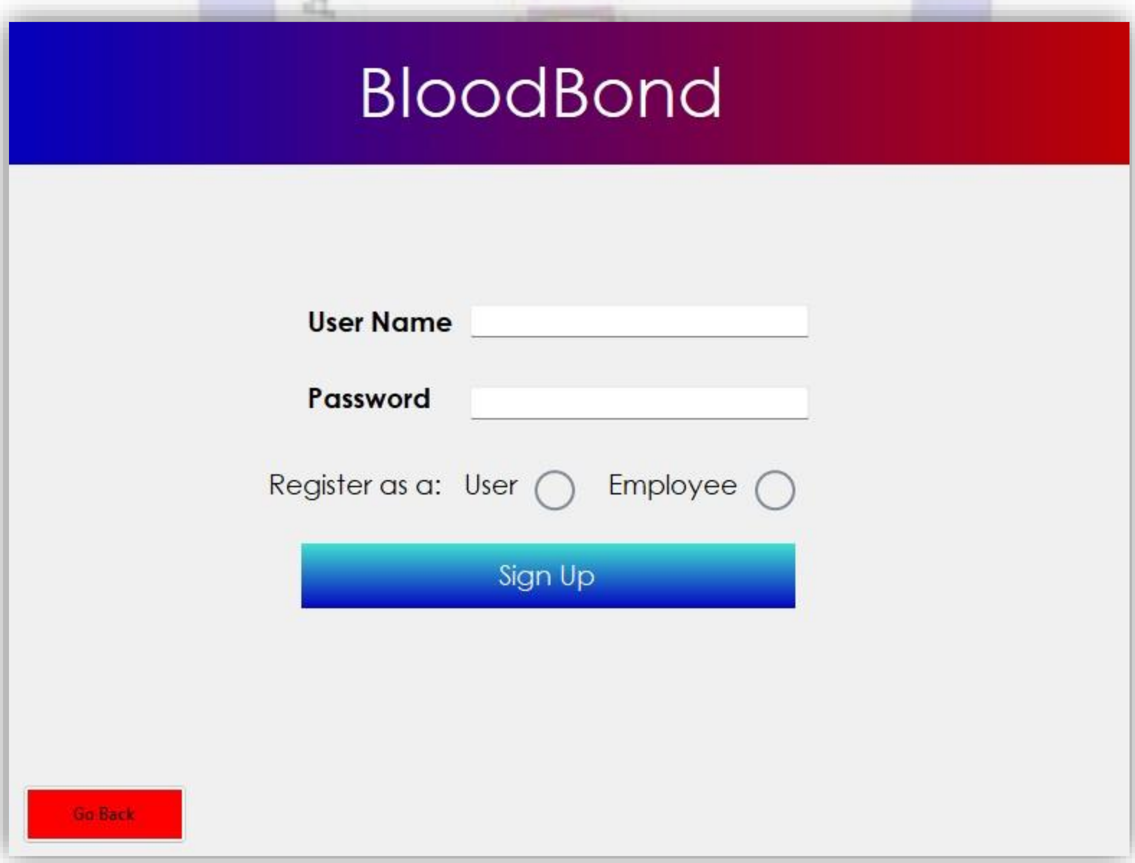
1. DNum
2. DName
3. DAge
4. DGender
5. DPhone
6. DAddress
7. DBGroup
8. DOrgan

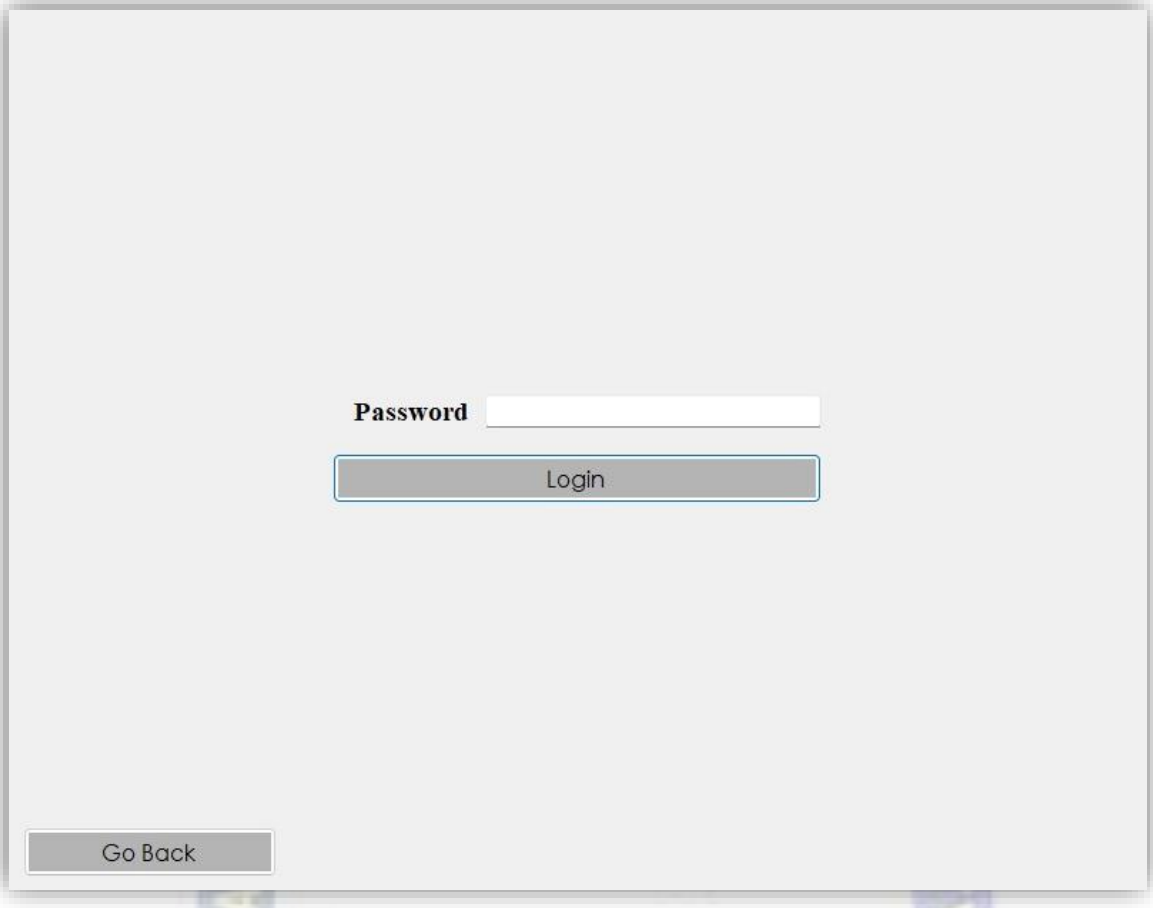
User Story of Recipient:

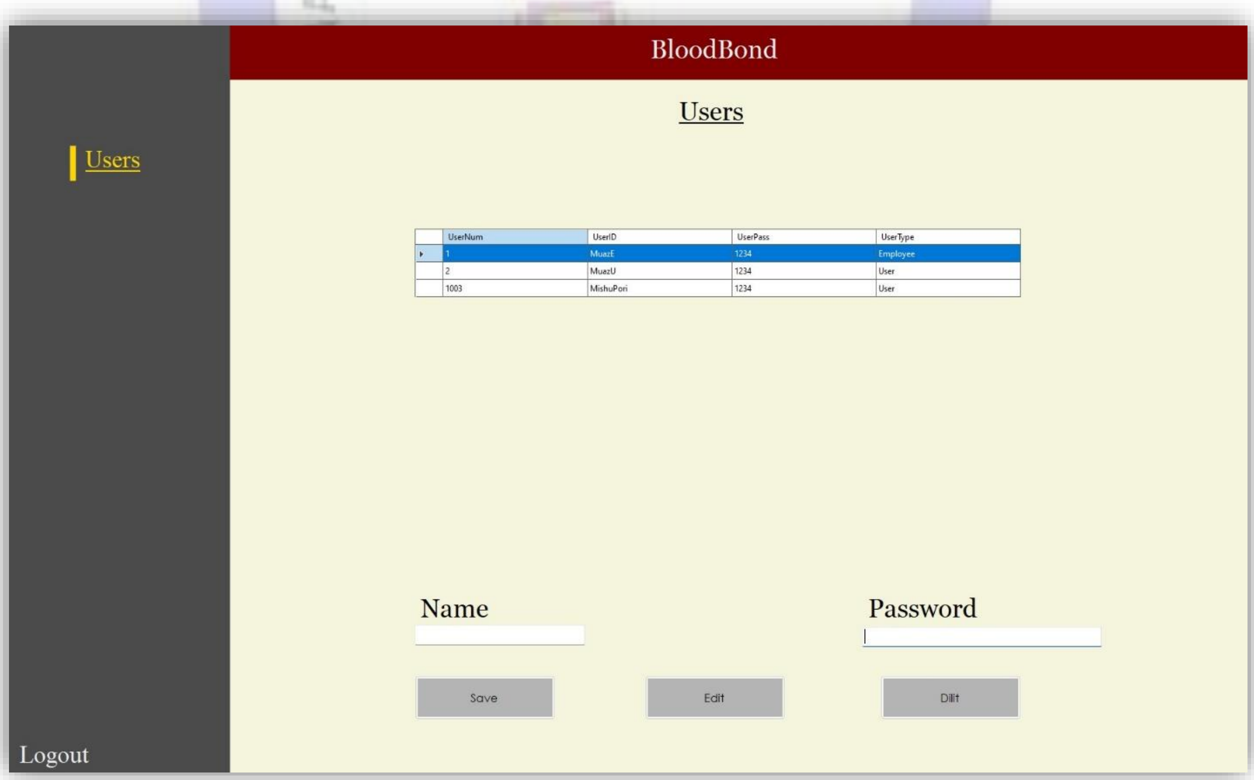
1. RNum
2. RName
3. RAge
4. RGender
5. RPhone
6. RAddress
7. RBGroup
8. ROrgan

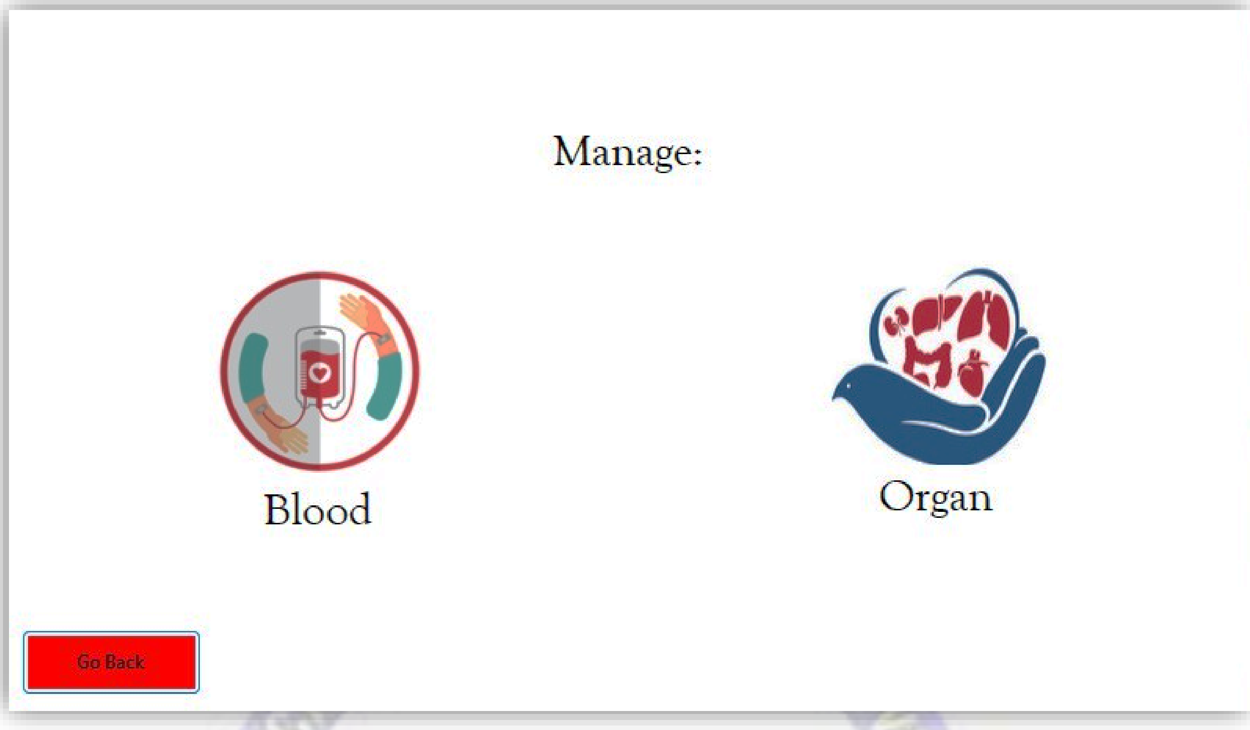
Form Images

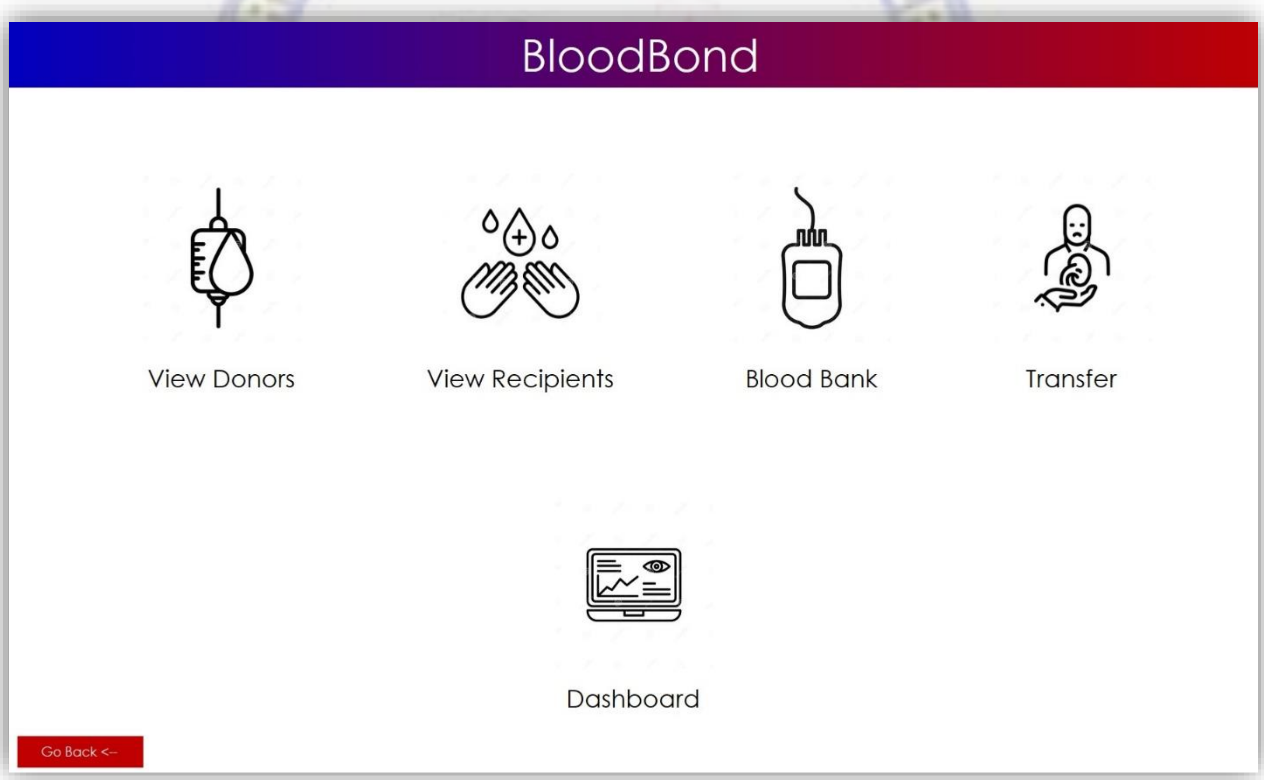

BloodBond
User Name
Password
Login
Don't Have an Account? Sign Up Now!
About
Continue as Admin
Exit

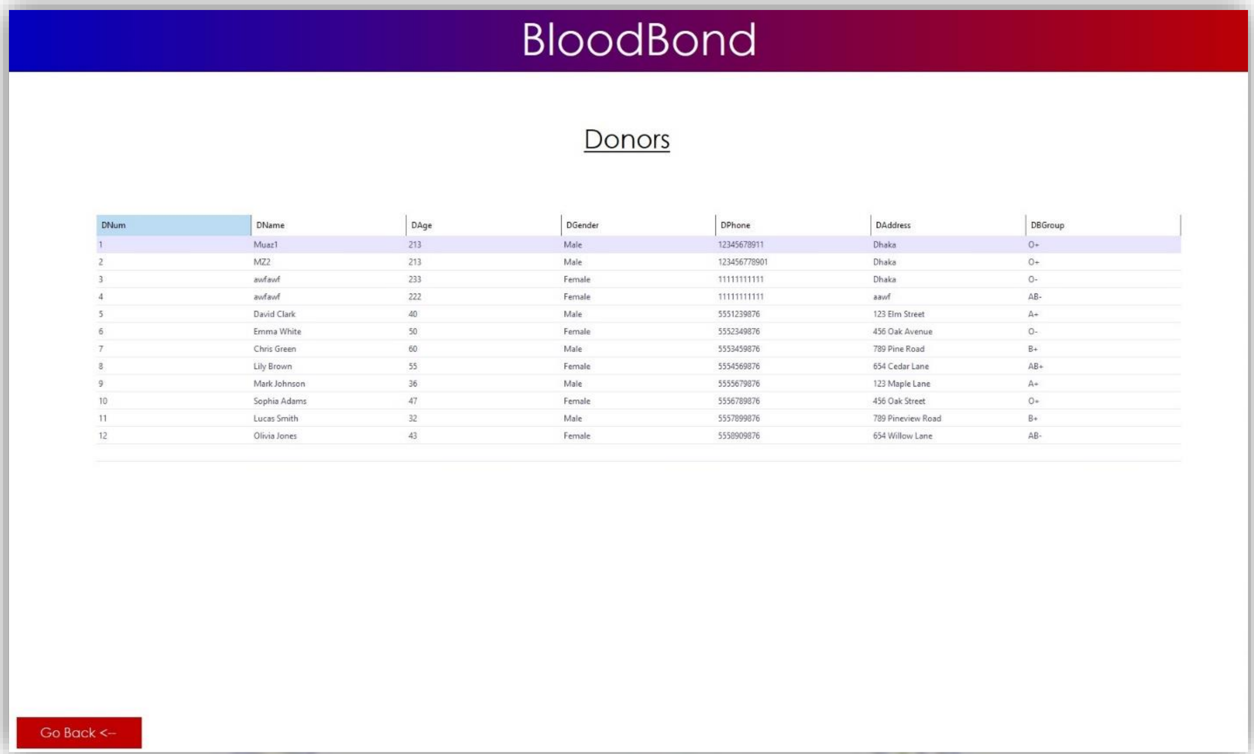

BloodBond
User Name
Password
Register as a: User Employee
Sign Up
Go Back

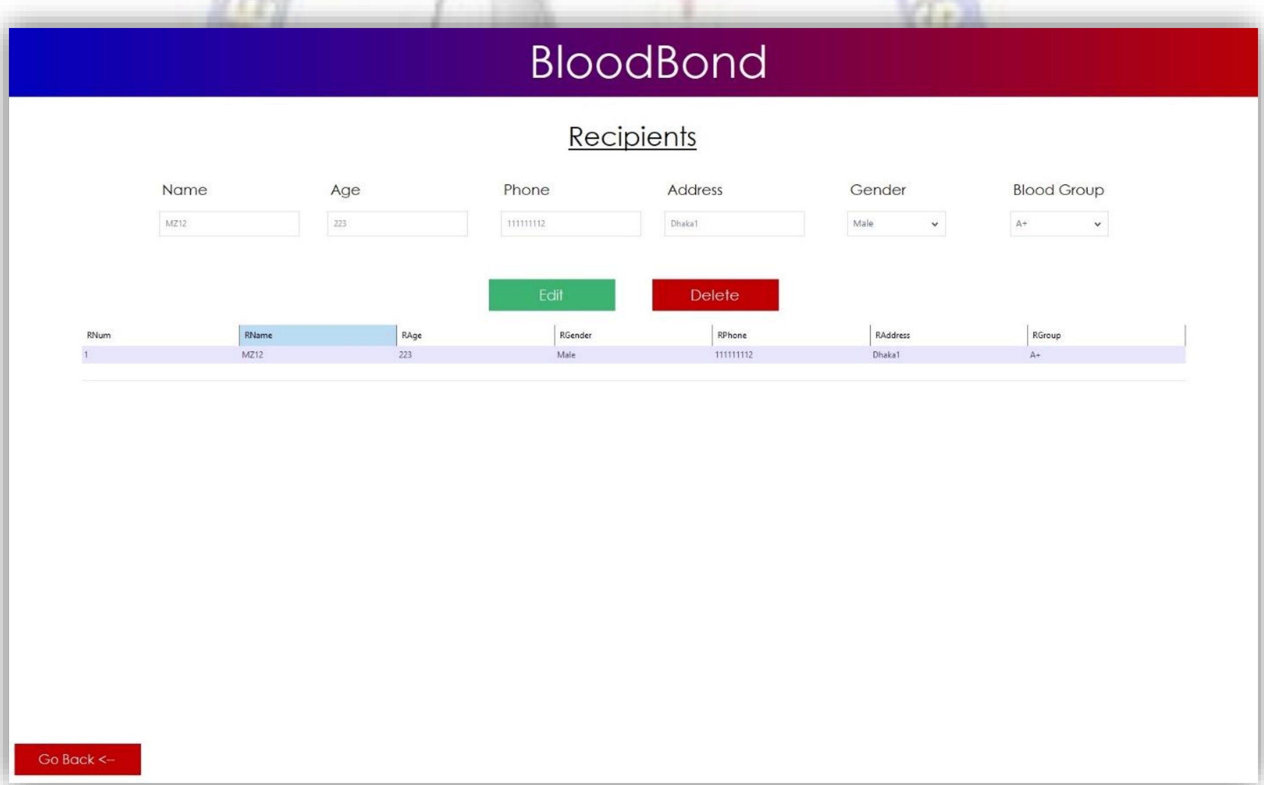

Password
Login
Go Back

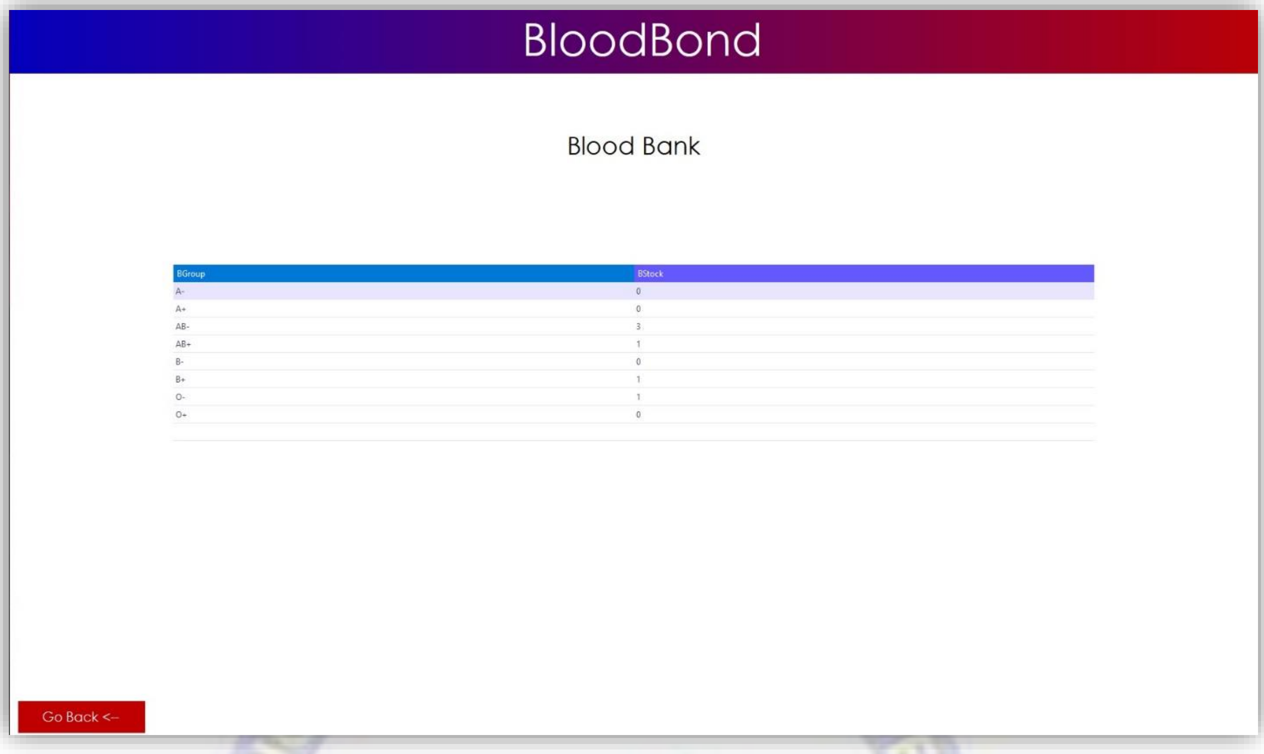

BloodBond
Users
Users
UserNum UserID UserPass UserType
1 MuazE 1234 Employee
2 MuazU 1234 User
1003 MishuPori 1234 User
Name
Password
Save
Edit
Dlit
Logout


Manage:
Blood
Organ
Go Back

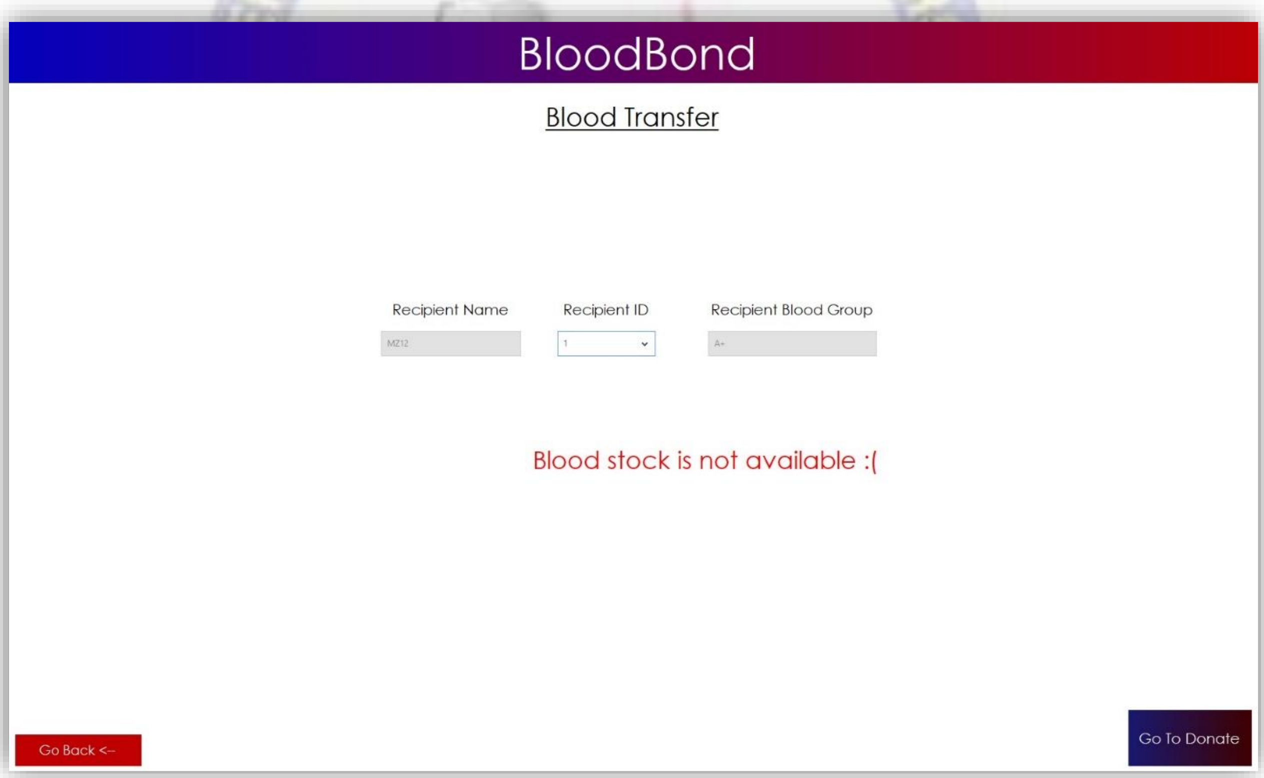

BloodBond
View Donors
View Recipients
Blood Bank
Transfer
Dashboard
Go Back <-

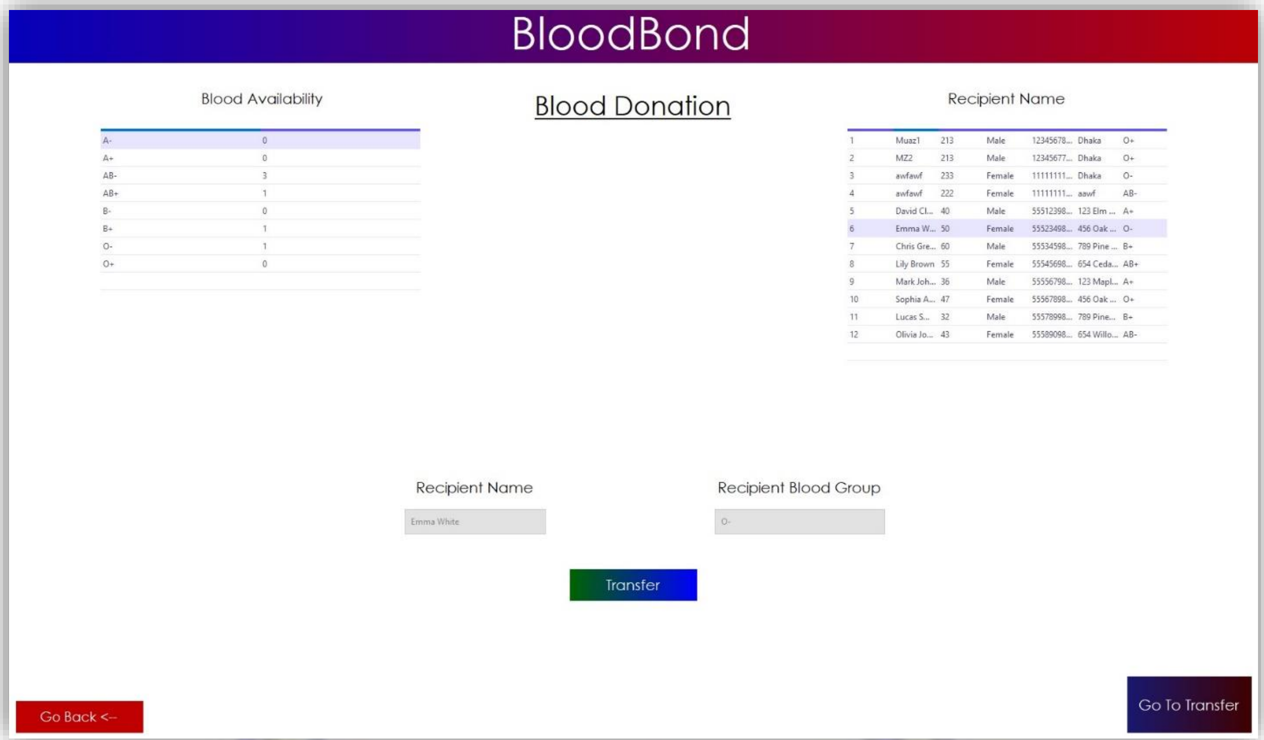

BloodBond
Donors
DNum DName DAge DGender DPhone DAddress DBGroup
1 Muaz1 213 Male 12345678911 Dhaka O+
2 MZ2 213 Male 123456778901 Dhaka O+
3 awfawf 233 Female 11111111111 Dhaka O-
4 awfawf 222 Female 11111111111 aawf AB-
5 David Clark 40 Male 5551239876 123 Elm Street A+
6 Emma White 50 Female 5552349876 456 Oak Avenue O-
7 Chris Green 60 Male 5553459876 789 Pine Road B+
8 Lily Brown 55 Female 5554569876 654 Cedar Lane AB+
9 Mark Johnson 36 Male 5555679876 123 Maple Lane A+
10 Sophia Adams 47 Female 5556789876 456 Oak Street O+
11 Lucas Smith 32 Male 5557899876 789 Pineview Road B+
12 Olivia Jones 43 Female 5558909876 654 Willow Lane AB-
Go Back <--

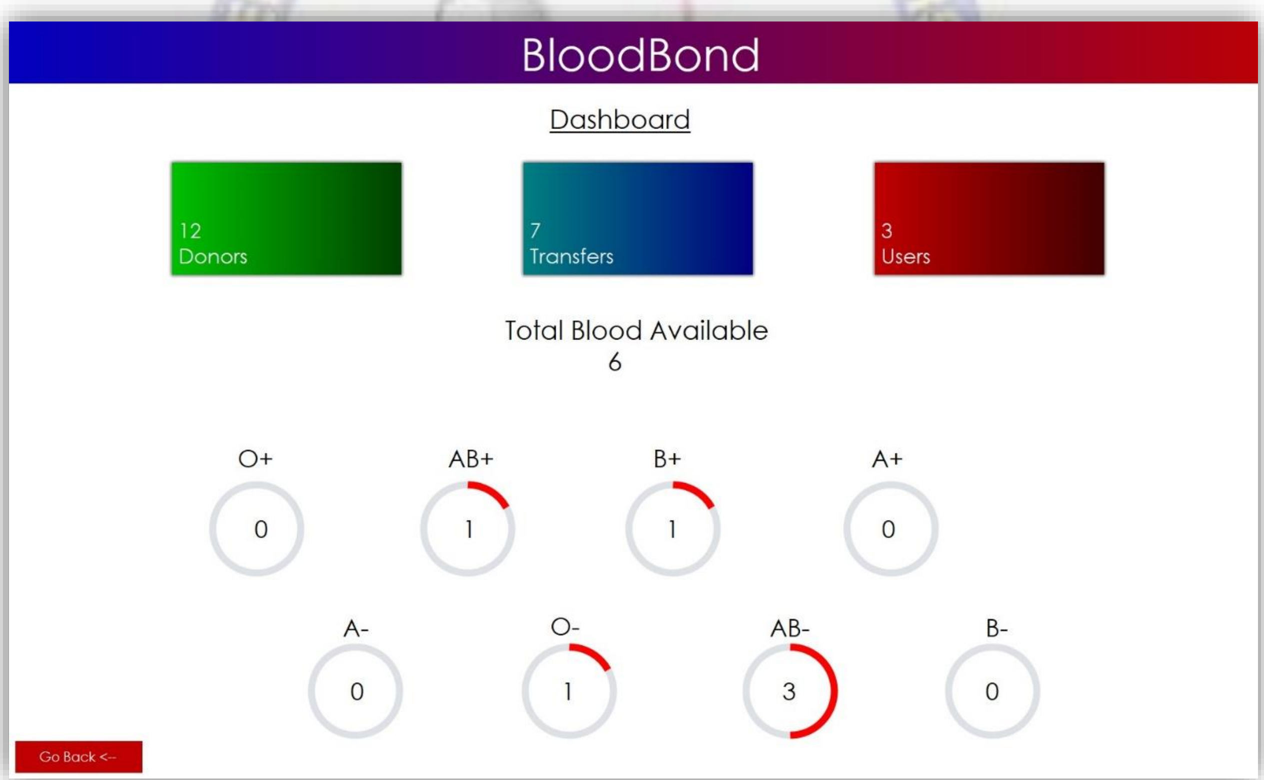

BloodBond
Recipients
Name Age Phone Address Gender Blood Group
MZ12 223 111111112 Dhaka1 Male A+
Edit
Delete
RNum RName RAge RGender RPhone RAddress RGroup
1 MZ12 223 Male 111111112 Dhaka1 A+
Go Back <--

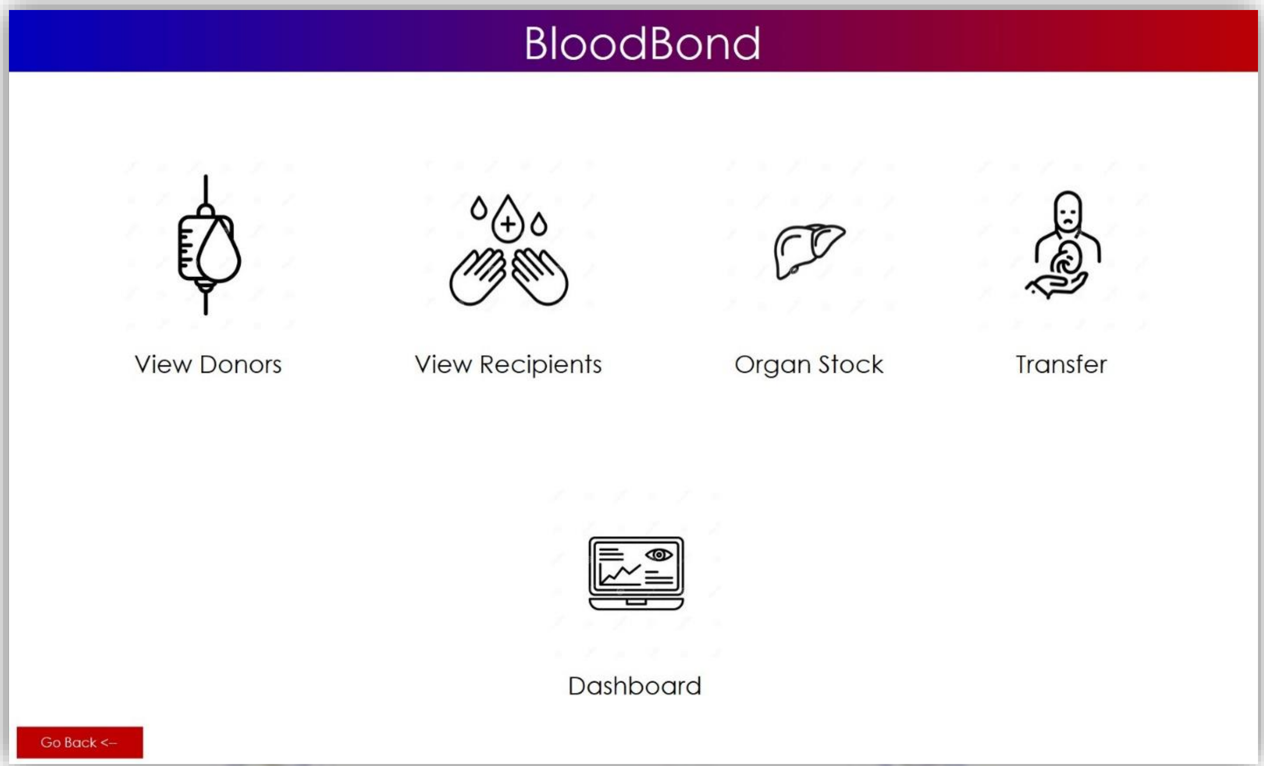


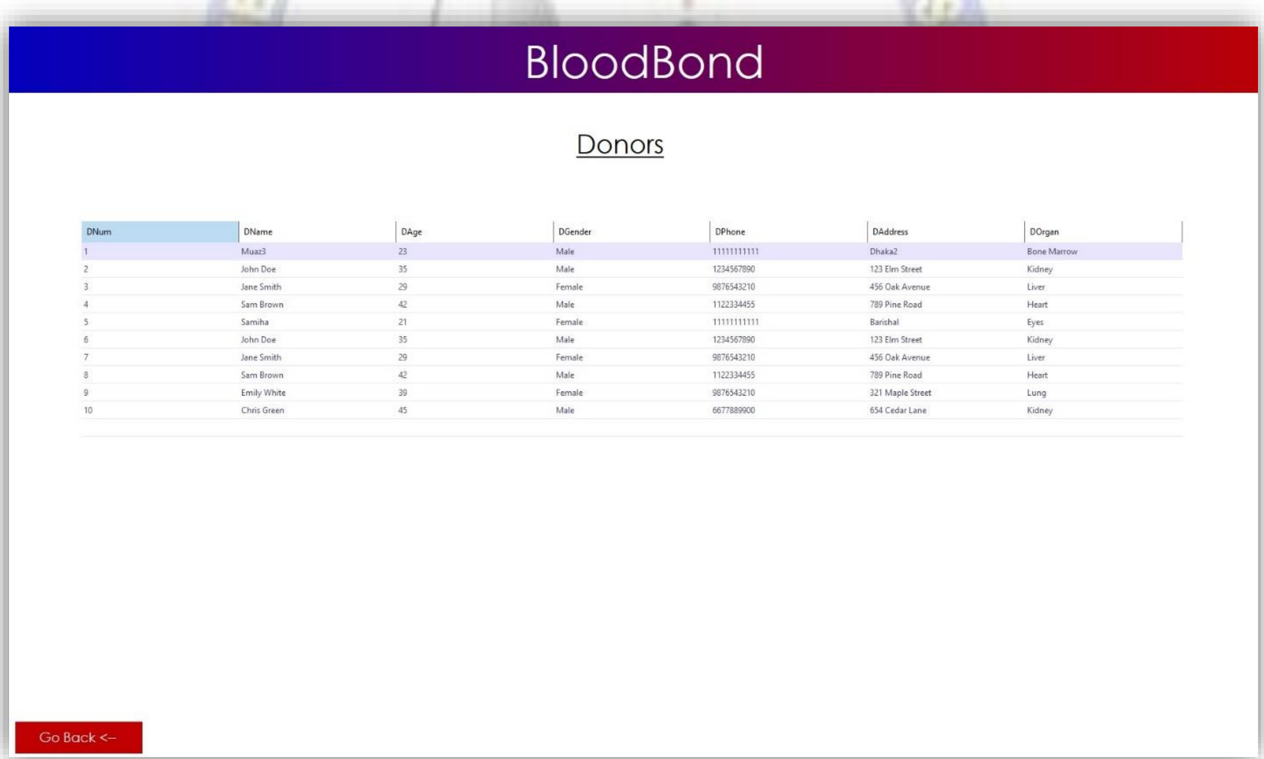
BGroup	BStock
A-	0
A+	0
AB-	3
AB+	1
B-	0
B+	1
O-	1
O+	0



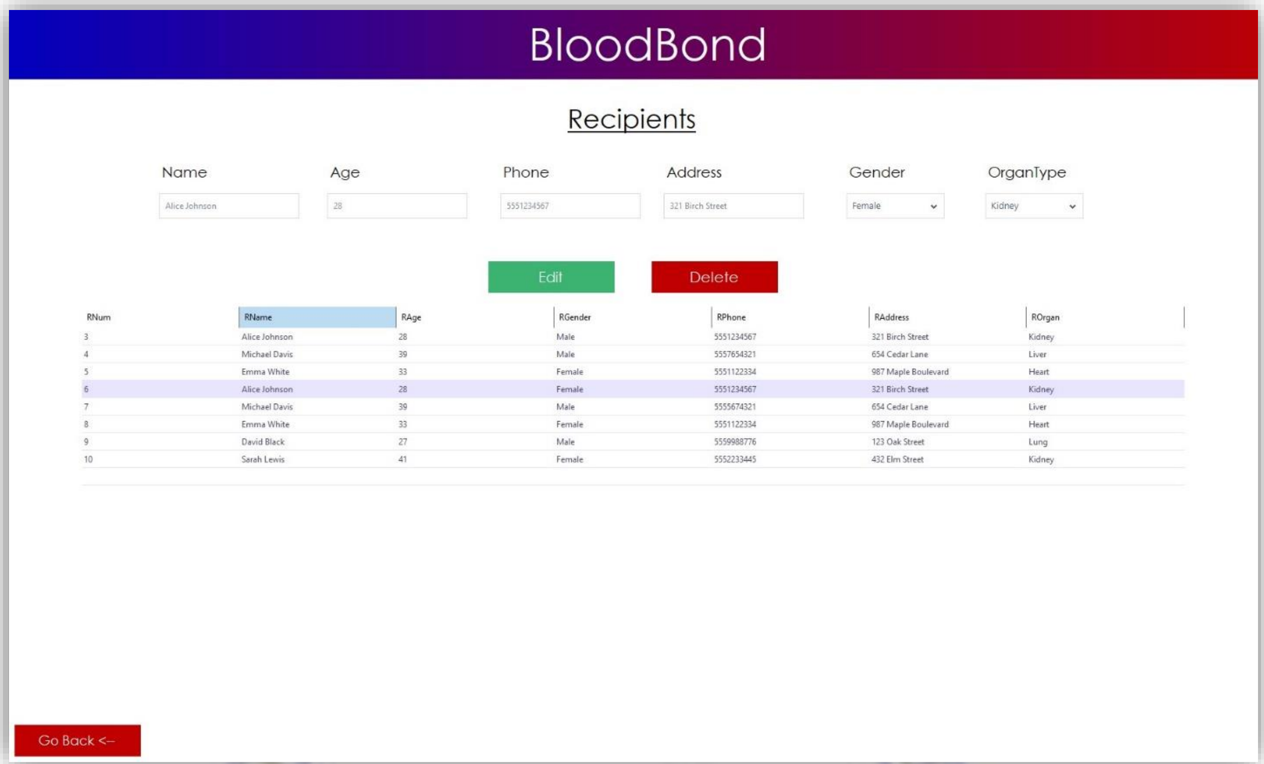

BloodBond
Blood Availability
A- 0
A+ 0
AB- 3
AB+ 1
B- 0
B+ 1
O- 1
O+ 0
Blood Donation
Recipient Name
1 Muaz1 213 Male 12345678... Dhaka O+
2 MZ2 213 Male 12345677... Dhaka O+
3 awfawf 233 Female 11111111... Dhaka O-
4 awfawf 222 Female 11111111... aawf AB-
5 David Cl... 40 Male 55512398... 123 Elm ... A+
6 Emma W... 50 Female 55523498... 456 Oak ... O-
7 Chris Gre... 60 Male 55534598... 789 Pine ... B+
8 Lily Brown 55 Female 55545698... 654 Ceda... AB+
9 Mark Joh... 36 Male 55556798... 123 Mapl... A+
10 Sophia A... 47 Female 55567898... 456 Oak ... O+
11 Lucas S... 32 Male 55578998... 789 Pine... B+
12 Olivia Jo... 43 Female 55589098... 654 Willo... AB-
Recipient Name
Emma White
Recipient Blood Group
O-
Transfer
Go Back <--
Go To Transfer


BloodBond
Dashboard
12
Donors
7
Transfers
3
Users
Total Blood Available
6
O+
0
AB+
1
B+
1
A+
0
A-
0
O-
1
AB-
3
B-
0
Go Back <--

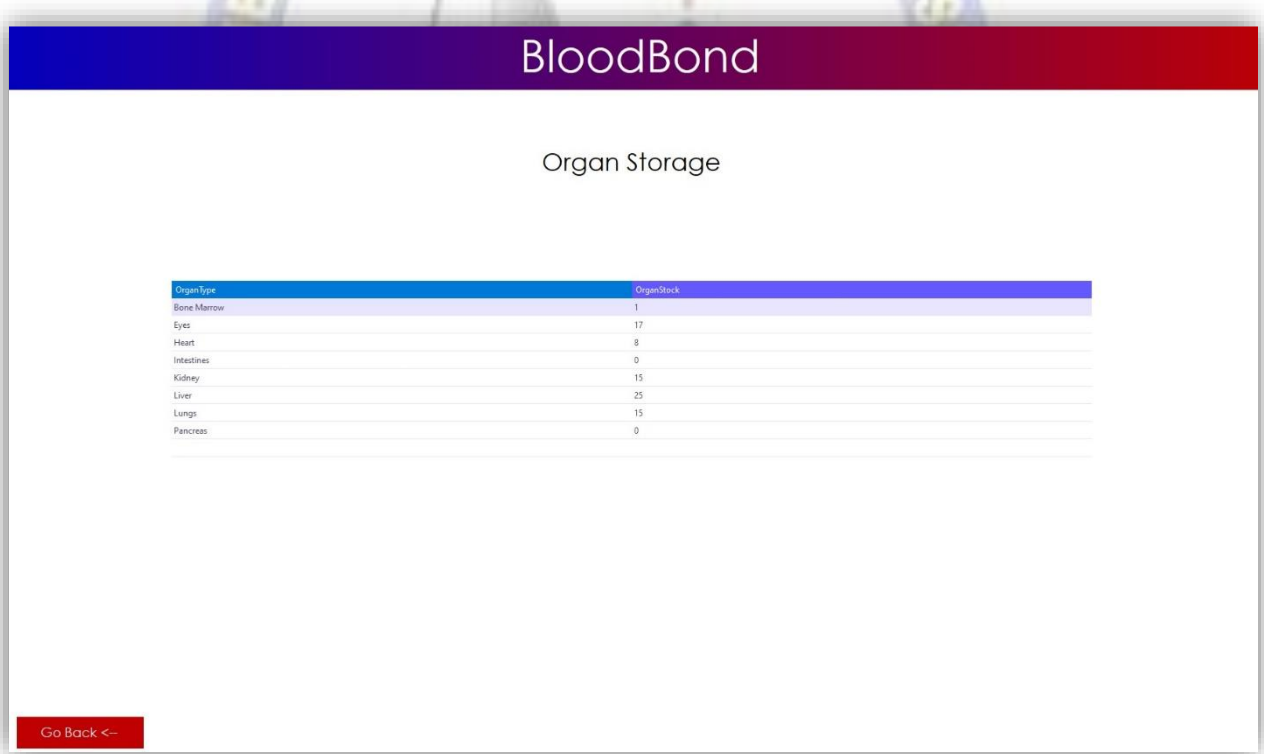




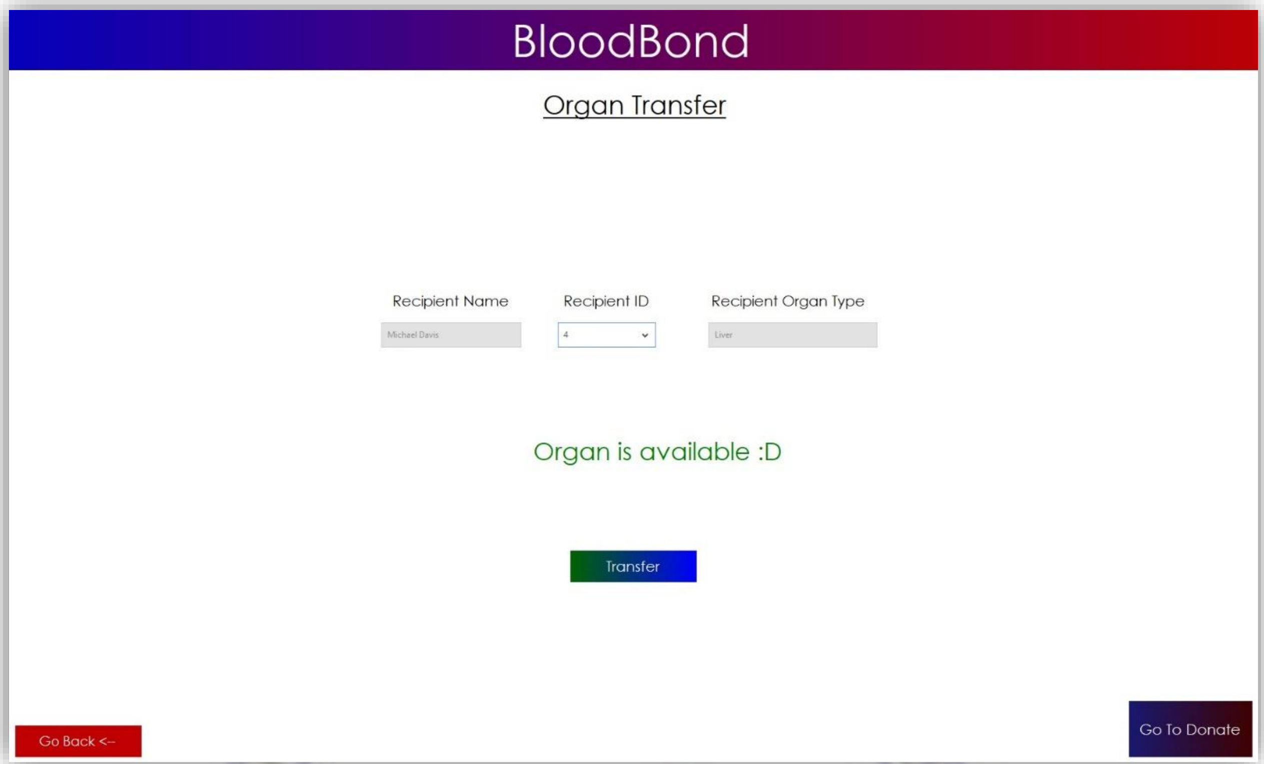
DNum	DName	DAge	DGender	DPhone	DAddress	DOrgan
1	Muaz3	23	Male	11111111111	Dhaka2	Bone Marrow
2	John Doe	35	Male	1234567890	123 Elm Street	Kidney
3	Jane Smith	29	Female	9876543210	456 Oak Avenue	Liver
4	Sam Brown	42	Male	1122334455	789 Pine Road	Heart
5	Samiha	21	Female	11111111111	Barishal	Eyes
6	John Doe	35	Male	1234567890	123 Elm Street	Kidney
7	Jane Smith	29	Female	9876543210	456 Oak Avenue	Liver
8	Sam Brown	42	Male	1122334455	789 Pine Road	Heart
9	Emily White	39	Female	9876543210	321 Maple Street	Lung
10	Chris Green	45	Male	6677889900	654 Cedar Lane	Kidney



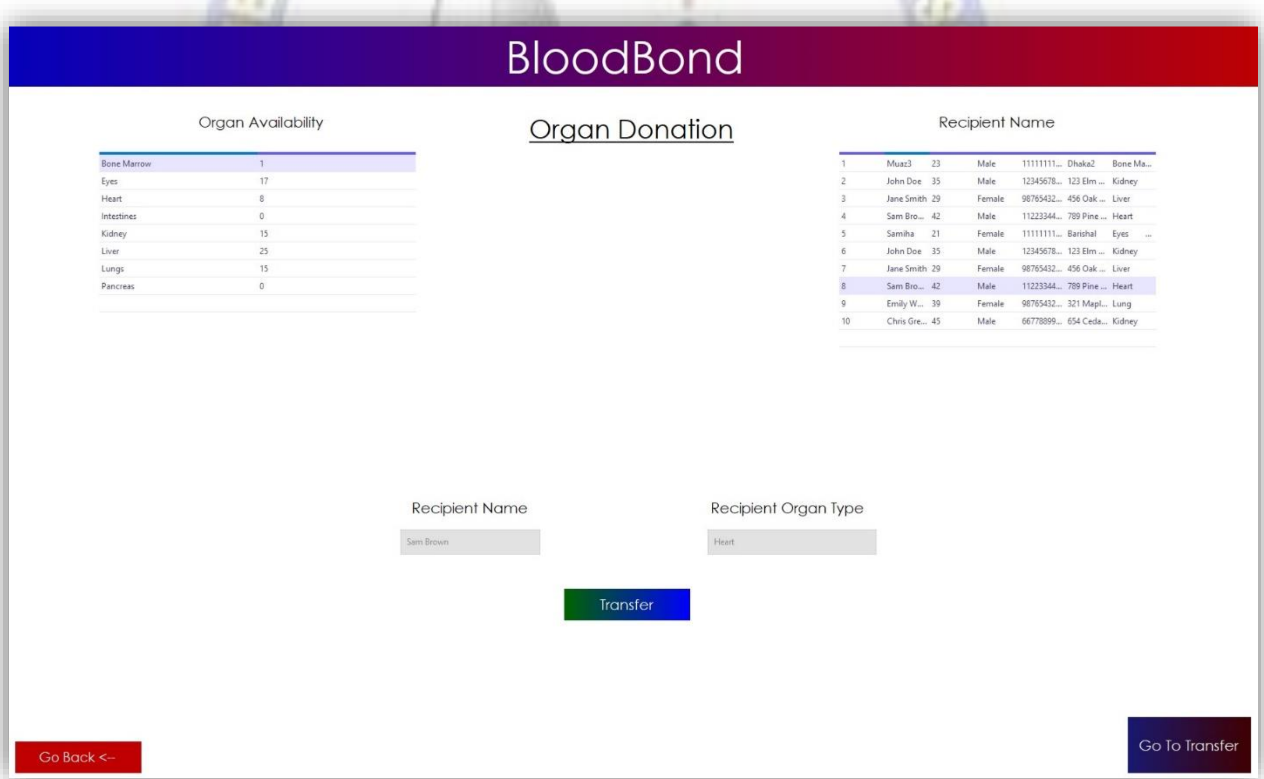
RNum	RName	RAge	RGender	RPhone	RAddress	ROrgan
3	Alice Johnson	28	Male	5551234567	321 Birch Street	Kidney
4	Michael Davis	39	Male	5557654321	654 Cedar Lane	Liver
5	Emma White	33	Female	5551122334	987 Maple Boulevard	Heart
6	Alice Johnson	28	Female	5551234567	321 Birch Street	Kidney
7	Michael Davis	39	Male	5555674321	654 Cedar Lane	Liver
8	Emma White	33	Female	5551122334	987 Maple Boulevard	Heart
9	David Black	27	Male	5559988776	123 Oak Street	Lung
10	Sarah Lewis	41	Female	5552233445	432 Elm Street	Kidney



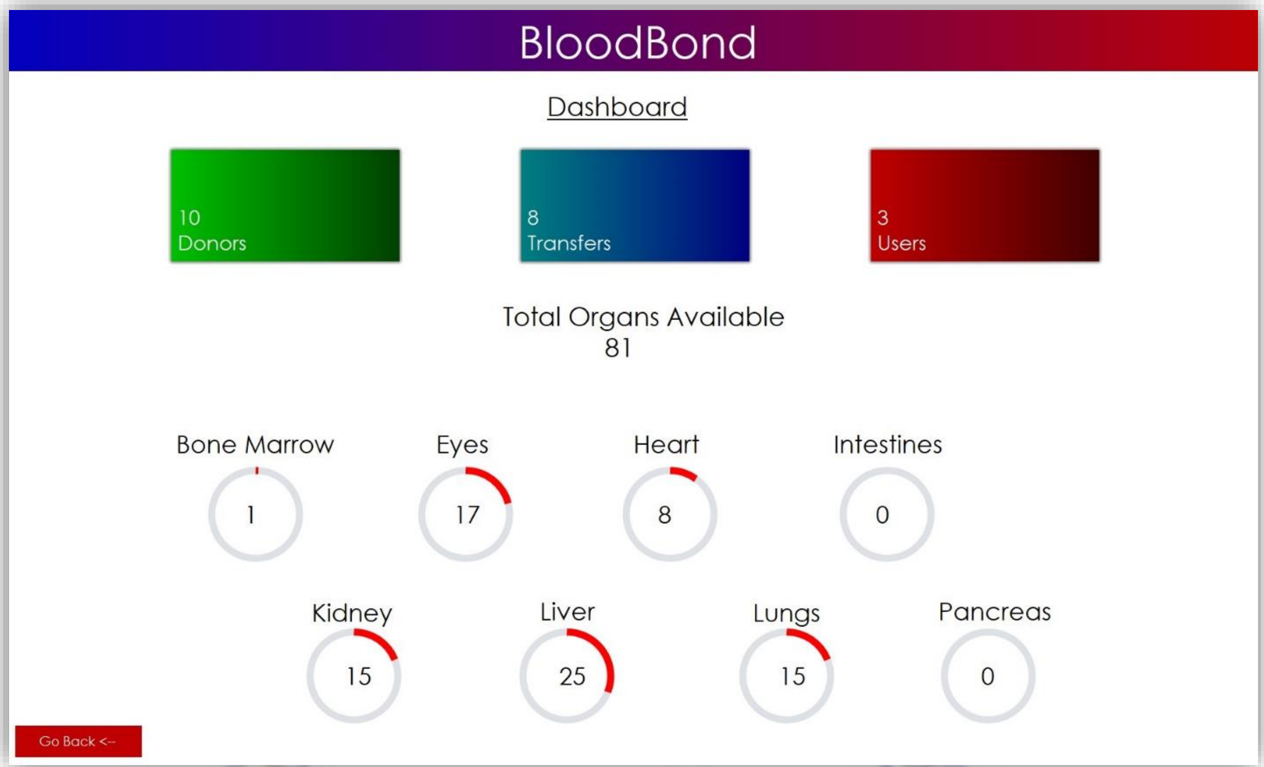
OrganType	OrganStock
Bone Marrow	1
Eyes	17
Heart	8
Intestines	0
Kidney	15
Liver	25
Lungs	15
Pancreas	0

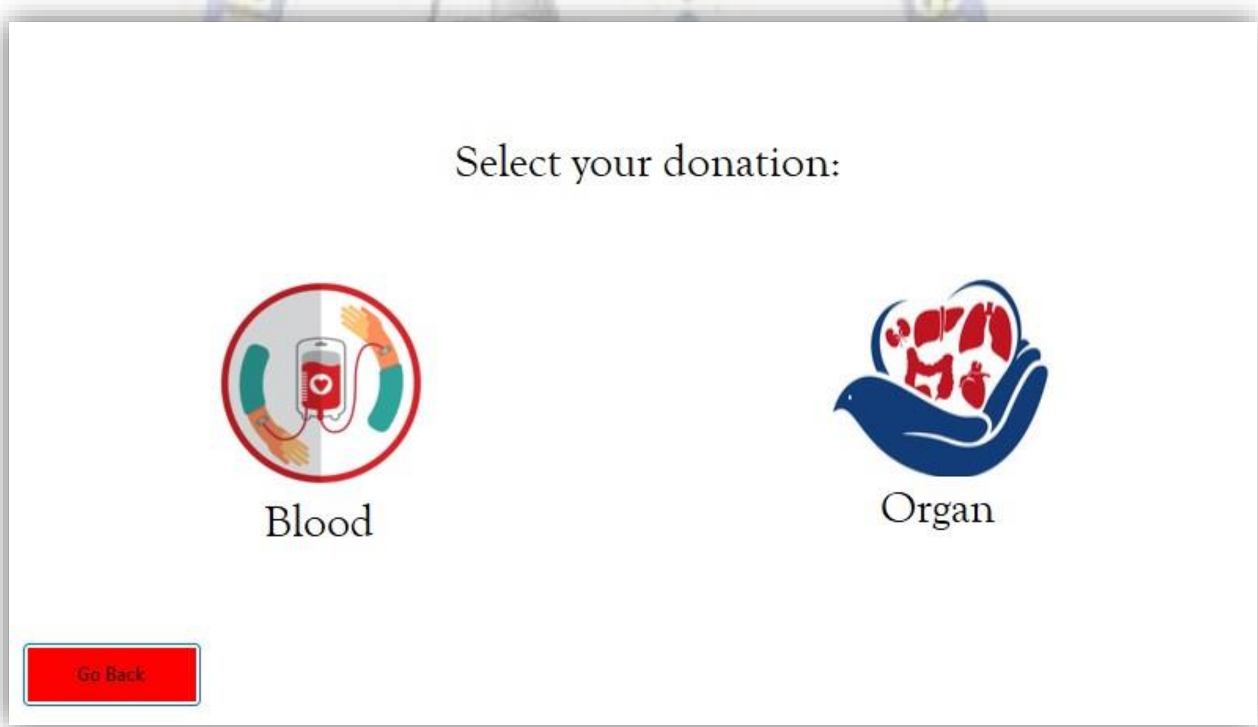


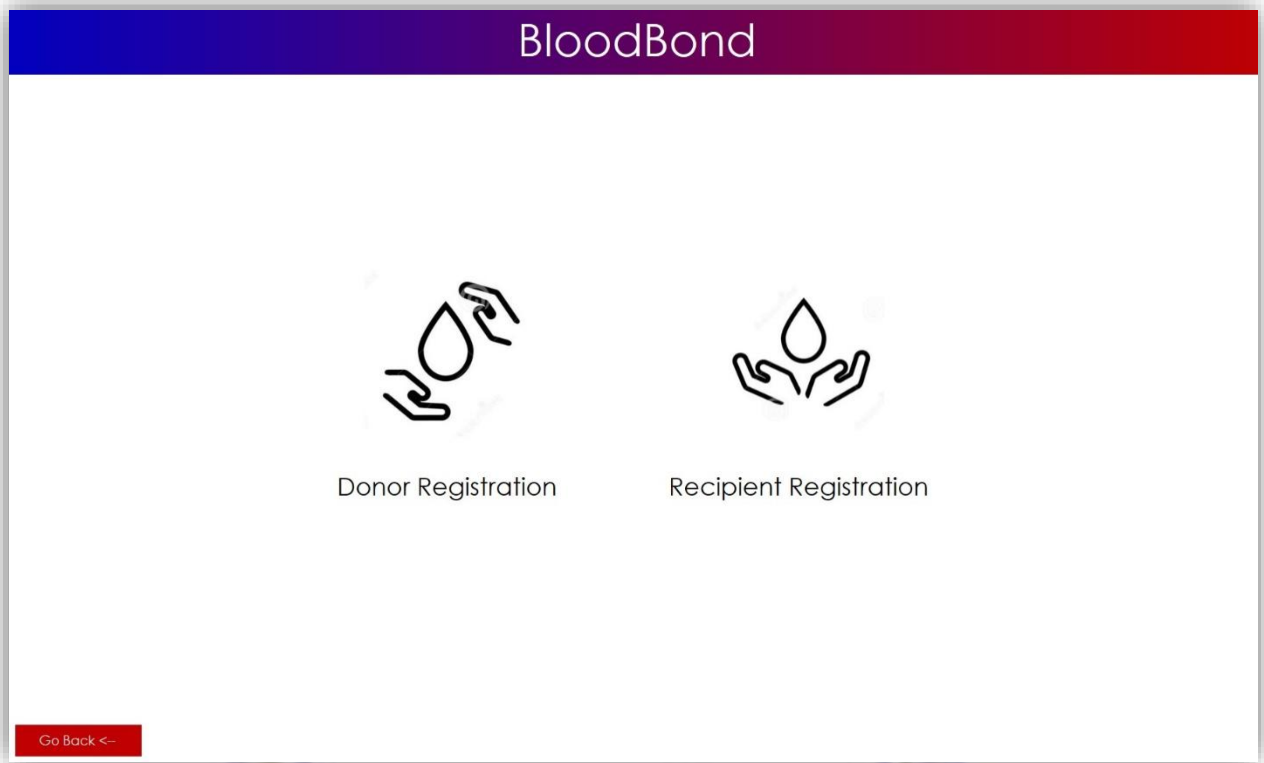
BloodBond
Organ Transfer
Recipient Name
Recipient ID
Recipient Organ Type
Michael Davis
4
Liver
Organ is available :D
Transfer
Go Back <--
Go To Donate

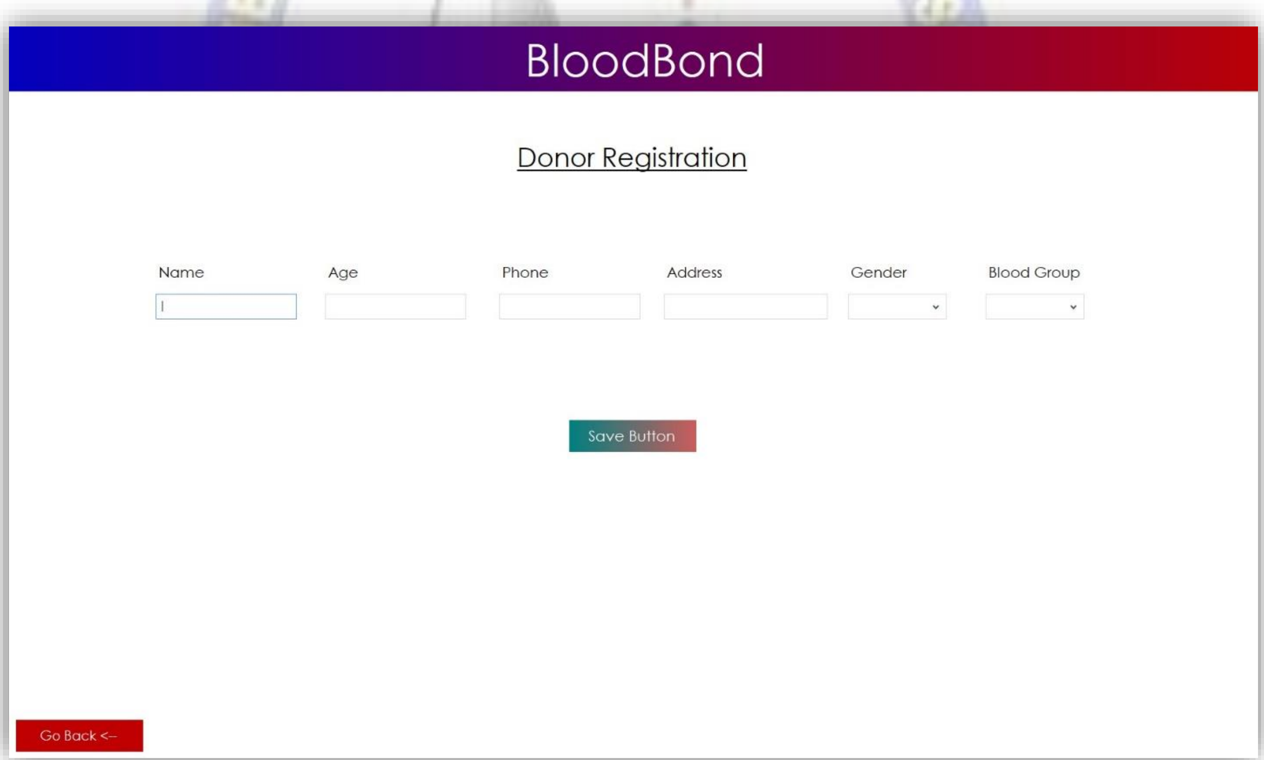


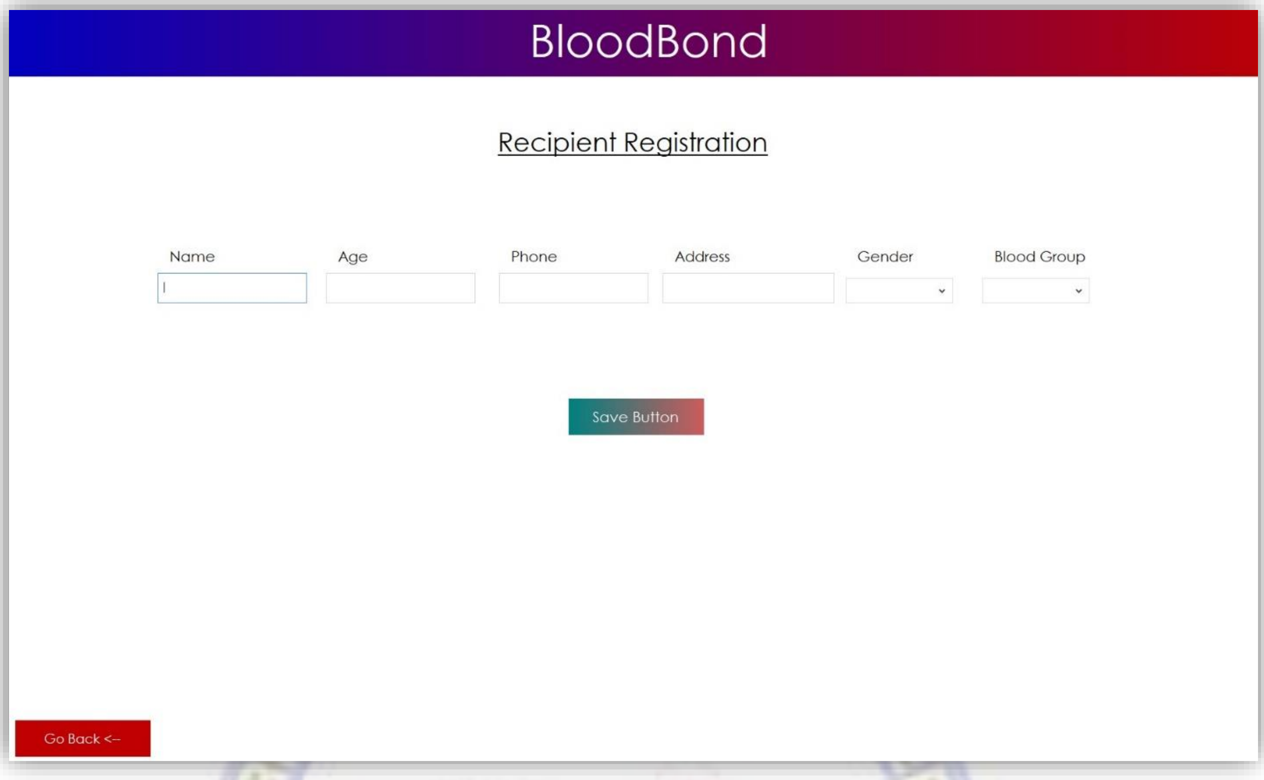
BloodBond
Organ Availability
Organ Donation
Recipient Name
Bone Marrow 1
Eyes 17
Heart 8
Intestines 0
Kidney 15
Liver 25
Lungs 15
Pancreas 0
1 Muaz3 23 Male 11111111... Dhaka2 Bone Ma...
2 John Doe 35 Male 12345678... 123 Elm ... Kidney
3 Jane Smith 29 Female 98765432... 456 Oak ... Liver
4 Sam Bro... 42 Male 11223344... 789 Pine ... Heart
5 Samiha 21 Female 11111111... Barishal Eyes ...
6 John Doe 35 Male 12345678... 123 Elm ... Kidney
7 Jane Smith 29 Female 98765432... 456 Oak ... Liver
8 Sam Bro... 42 Male 11223344... 789 Pine ... Heart
9 Emily W... 39 Female 98765432... 321 Mapl... Lung
10 Chris Gre... 45 Male 66778899... 654 Ceda... Kidney
Recipient Name
Recipient Organ Type
Sam Brown
Heart
Transfer
Go Back <--
Go To Transfer

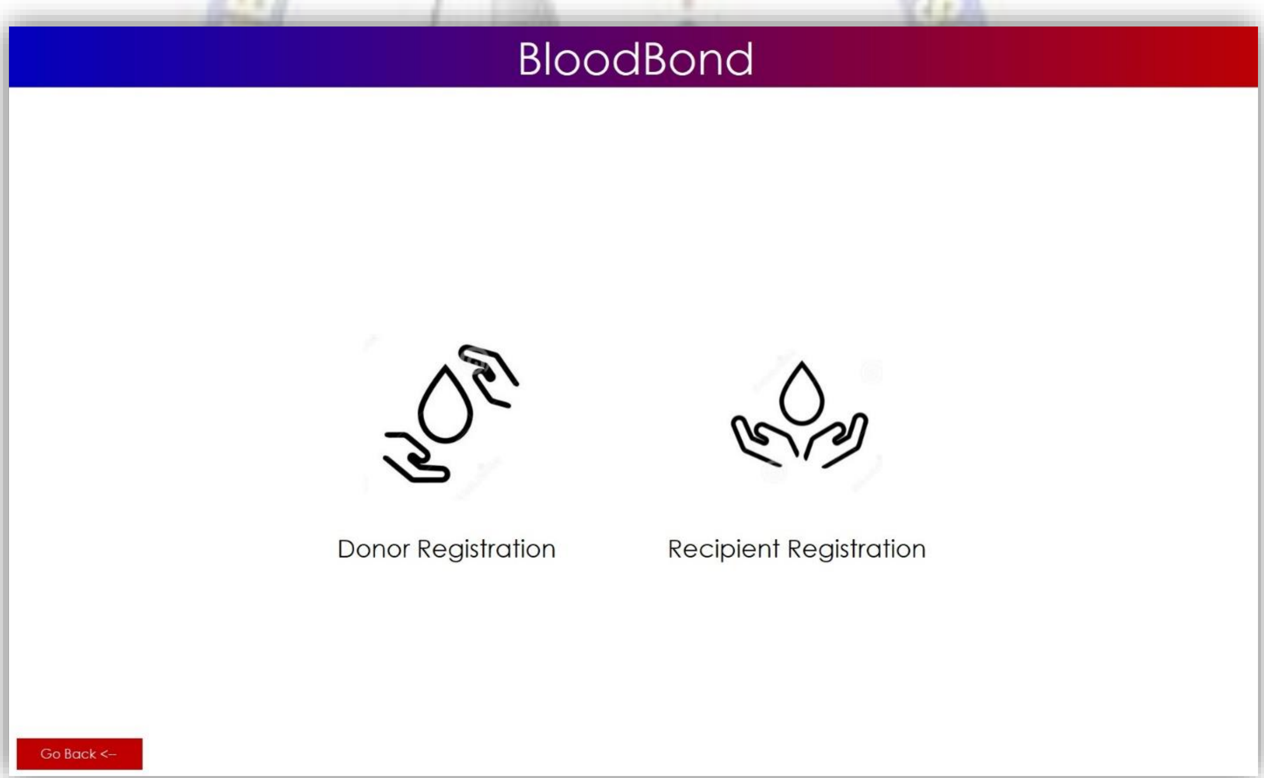

BloodBond
Dashboard
10
Donors
8
Transfers
3
Users
Total Organs Available
81
Bone Marrow
1
Eyes
17
Heart
8
Intestines
0
Kidney
15
Liver
25
Lungs
15
Pancreas
0
Go Back <--

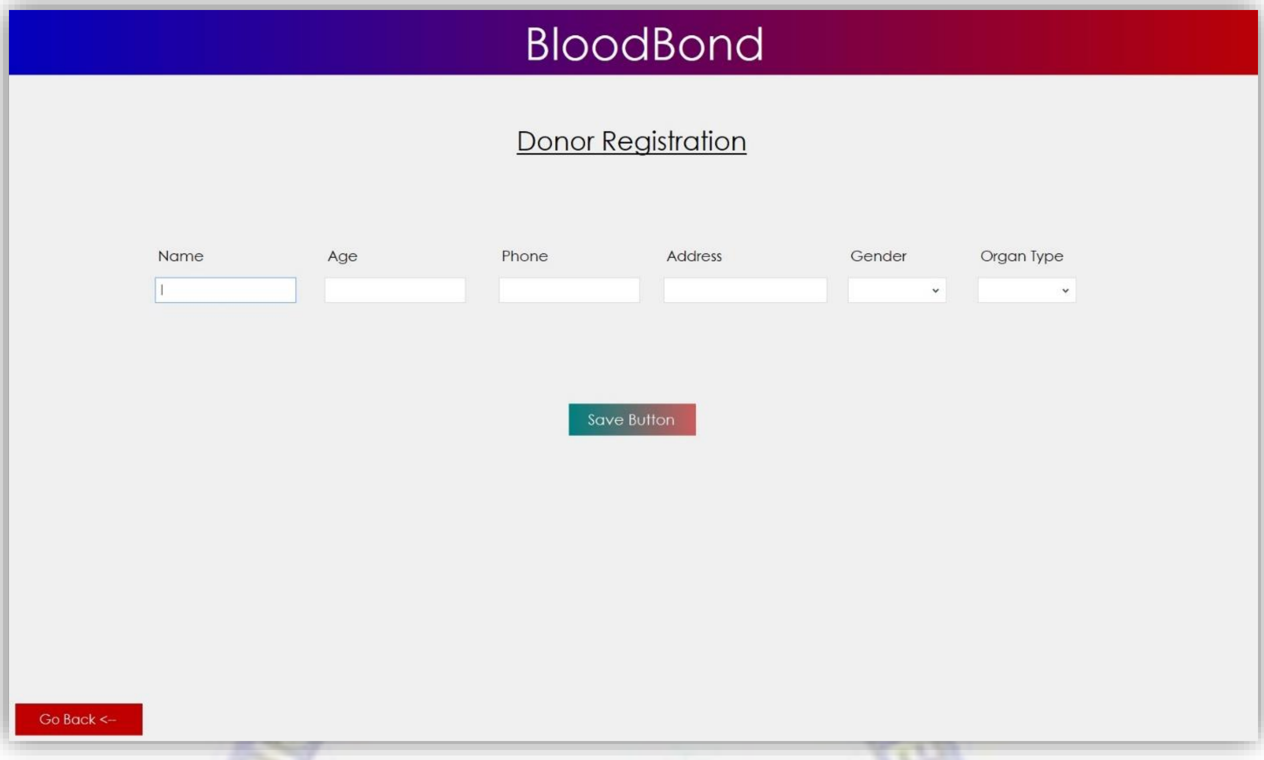

Select your donation:
Blood
Organ
Go Back

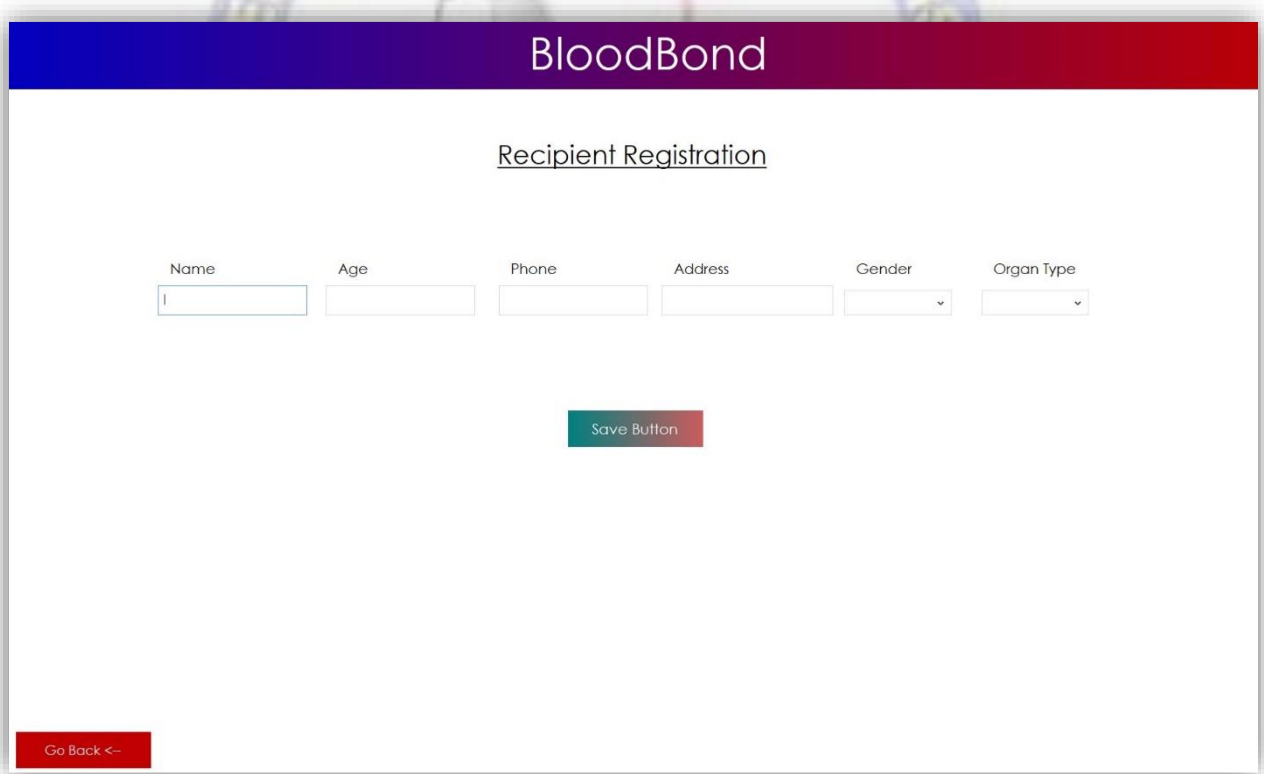

BloodBond
Donor Registration
Recipient Registration
Go Back <--


BloodBond
Donor Registration
Name
Age
Phone
Address
Gender
Blood Group
Save Button
Go Back <--

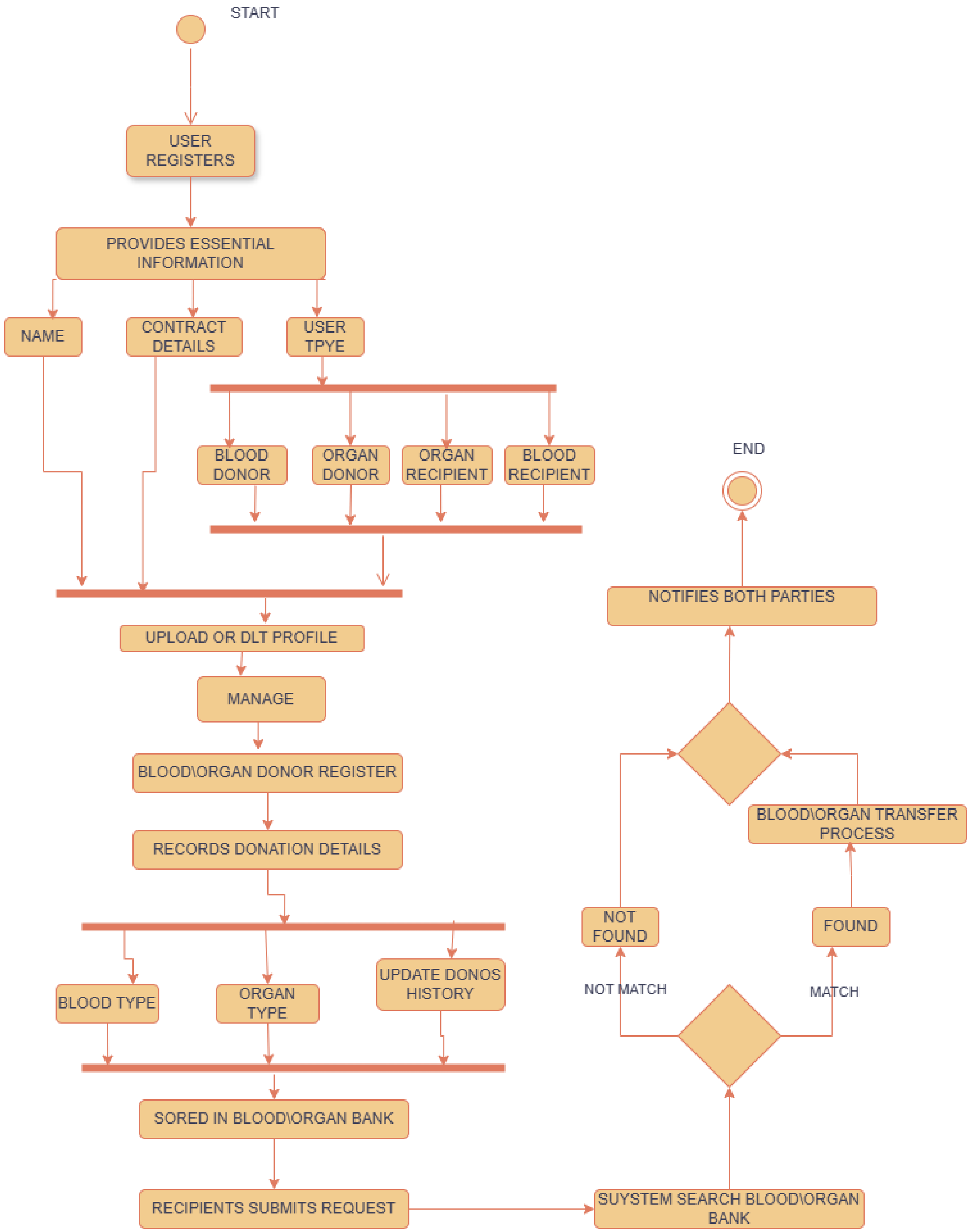

BloodBond
Recipient Registration
Name
Age
Phone
Address
Gender
Blood Group
Save Button
Go Back <--


BloodBond
Donor Registration
Recipient Registration
Go Back <--

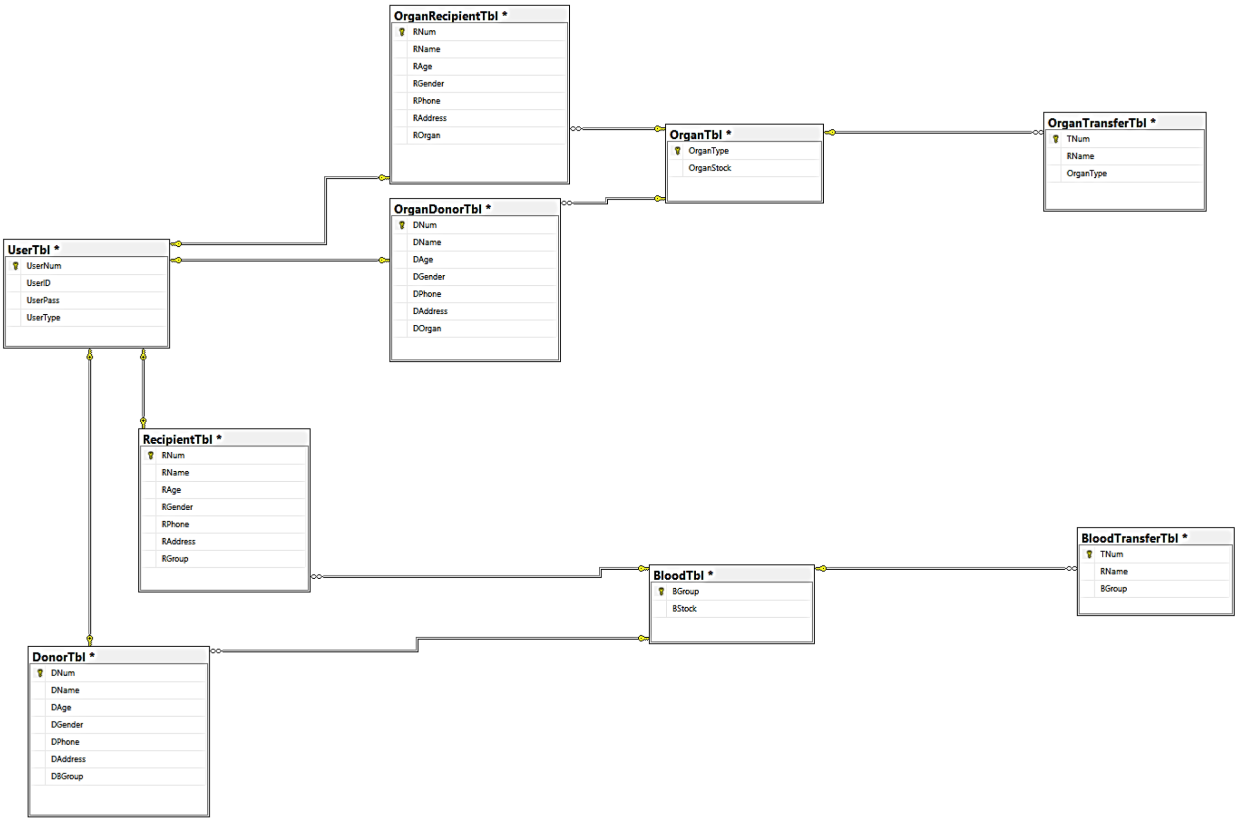

BloodBond
Donor Registration
Name
Age
Phone
Address
Gender
Organ Type
Save Button
Go Back <--


BloodBond
Recipient Registration
Name
Age
Phone
Address
Gender
Organ Type
Save Button
Go Back <--

Activity Diagram



Schema Diagram:



Conclusion:

The Blood and Organ Donation System project aims to enhance the efficiency and reliability of the donation process by providing a streamlined platform for donors, recipients, and healthcare providers. Key features include secure data management, ensuring timely and accurate donor-recipient matching. The system also offers analytical tools to predict supply and demand trends, improving resource allocation. By prioritizing security and user-friendly access, this project supports a more organized and effective donation process, ultimately contributing to better healthcare outcomes and saving more lives.